Game 3 Catch the Fruit

Before you start

These are teacher-designed Scratch 3 instructions. Work through the core game first, then test it with a partner. Save your project in the class folder.

Build or open the starter

Choose a basket and one fruit sprite. Position the basket near the bottom. Use a simple backdrop. The teacher may provide the falling-fruit script as a starter.

Select the basket and add:

```
when green flag clicked
go to x: (0) y: (-140)

when [right arrow] key pressed
change x by (15)

when [left arrow] key pressed
change x by (-15)
```

Keep the basket inside the stage during testing. Optional teacher support: limit x to -200 through 200.

Make the fruit fall and reset

Select the fruit. Build this script, using the basket's actual sprite name:

```
when green flag clicked
go to x: (pick random (-200) to (200)) y: (160)
forever
  change y by (-4)
  if <touching [Basket] ?> then
    play sound [pop] until done
    go to x: (pick random (-200) to (200)) y: (160)
  end
  if <(y position) < (-170)> then
    go to x: (pick random (-200) to (200)) y: (160)
  end
  wait (0.03) seconds
end
```

Test and improve

Test a catch and a miss. Both should send the fruit back to the top so play continues.

If fruit falls too fast, change -4 to -2. If a catch makes no sound, check the touching block and that both sprites can meet.

Optional score: reset Score to 0 when the game starts and add 1 only inside the touching-basket branch.

Explain

Point to the repeated falling action, the catch check and the reset after a miss.

Sources and related skills

Related book pages are included after this guide. They teach supporting skills; this game is a separate classroom project.

Official Scratch project ideas: https://scratch.mit.edu/ideas

4.2 Command your sprite

In this lesson

You will learn:

➔ how to add commands to your program.

Load program file

Last lesson you saved your work as a file. Now you will load the file. Load means get a file from storage.



Open the file menu. Click on 'load from your computer'. Choose the file you made last time.

Coloured dots and blocks

You will make a Scratch program. A program is made of **commands**. Commands tell the computer what to do.

Scratch commands are blocks. You fit the blocks together. That makes the program.

Different types of blocks are different colours.

- 'Motion' blocks are blue.
- 'Sound' blocks are pink.

Look for the coloured dots on the left of the screen. Choose a colour to help you pick the blocks you need.

Actions

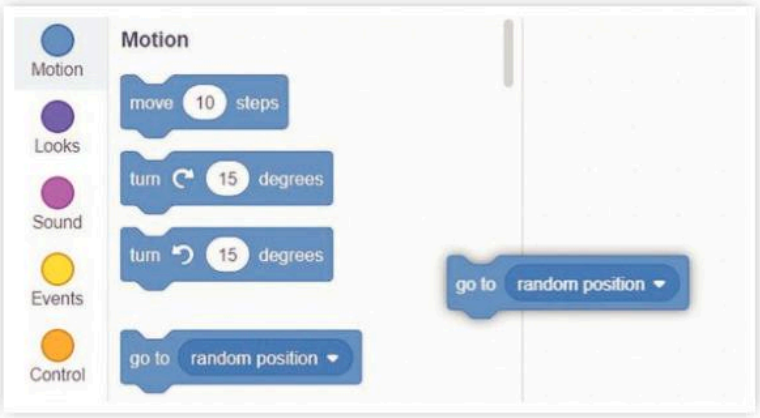
Choose a block. It will make your sprite move.

Click on the blue dot. You will see 'Motion' blocks.

Find the block that says 'go to random position'.

Drag this block to the middle of the screen.

Click on the block. The sprite will move.



Sounds

Choose a block. It will play a sound. Click on the pink dot. You will see 'Sound' blocks. Choose the 'play sound' block.

Now you have two blocks. Fit the blocks together.

Click on the two blocks. What happens?





Activity

Make the two-block program shown on this page.



Extra challenge

Add any other block to the program. See what happens when you run the program.

Think again

Say what blocks you used. Say what each block did.

4.3 A start event

In this lesson

You will learn:

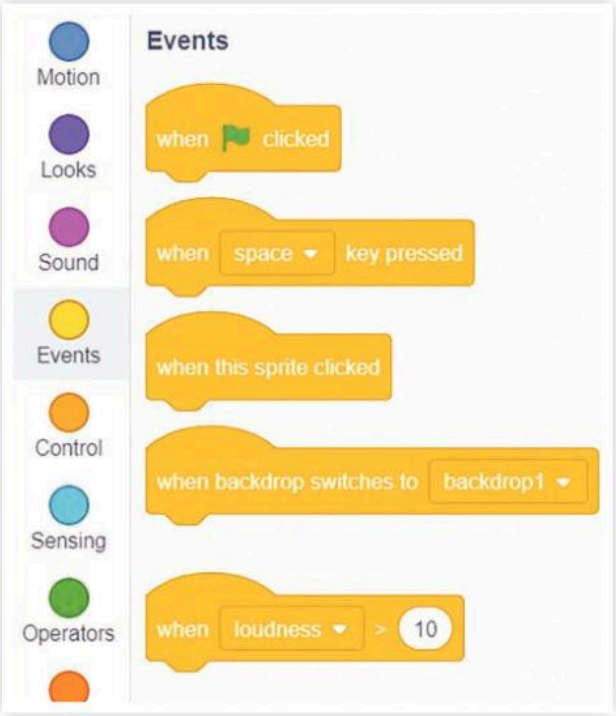
- to choose an event that starts the program.

Events

An **event** means anything that happens. In Scratch 'Events' are things the **user** does. The user is the person who uses the program. Now you will choose the 'Event' that lets the user start your program.

Click on the yellow dot that says 'Events'.

What 'Event' blocks can you see? What does each one say?

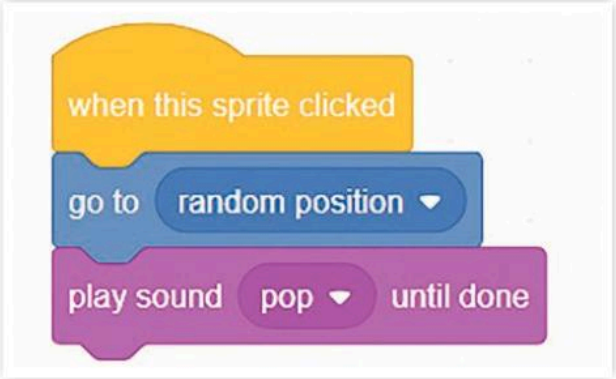


Choose an event

To choose an 'Event' drag the block to the program area.

Find the 'Event' block that says 'when this sprite clicked'.

Fit the program blocks onto the 'Event' block. Now the event will start the program.



Change colour

Now you will make the sprite change colour. Click on the purple dot that says 'Looks'. These command blocks change the look of the sprite.

Find the block that says 'change color effect by…'.

Drag this block to the script area and add it to the program.





Activity

Change your program.

- Add the 'Event' block which says 'when this sprite clicked'.
- Add the 'Looks' block which changes the colour of the sprite.

Click the sprite to make it move.

Save the file.



Extra challenge

Choose a different start event. Change the program to the new event.

Think again

Draw the 'Event' block you used. Say what it does.

If you used more than one, draw them all.

4.4 Forever loop

In this lesson

You will learn:

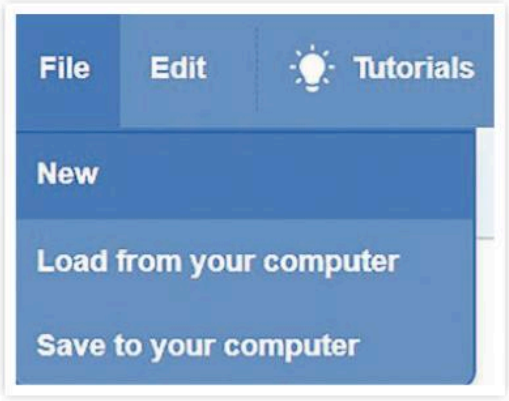
- how to change the program so the sprite moves 'forever'.

Last lesson you wrote a program. The program started when you clicked on the sprite. The sprite moved to a new place. The sprite made a sound. These things happened one time.

Now you will make a new program. In this program the sprite will move over and over again. The action will repeat.

New program

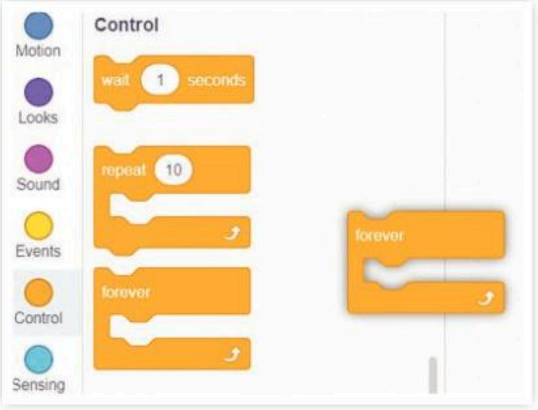
Open the file menu. Click on 'New' to make a new program.



Control blocks

Click on the orange button. You will see the orange 'Control' blocks. Find the block that says 'forever'. This block is called the **forever loop**. Drag it to the middle of the screen.

A forever loop is a way to control the program. Commands inside the forever loop will repeat.



Add commands

Here are the commands you want to repeat:

- Make the sound 'pop'.
- Go to a random position.

Put these blocks inside the forever loop. They are the same blocks you used in the last program.

Wait

Add one more block inside the forever loop. You will find it with the orange control blocks. It says 'wait 1 seconds'. This will make the sprite wait for 1 second between jumps.

The completed program looks like this.





Activity

Start a new program. Choose a sprite and a background.

Make the program shown on this page. Run the program to see what happens.

Think again

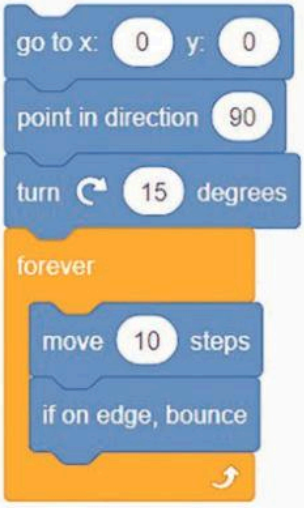
Say what happens when you put blocks inside a forever loop.



Extra challenge

Here is a new program that uses the forever loop.

- Make this program.
- Add a start 'Event' to the program.
- Change the numbers in the program and see what happens.



4.6 Find and fix

In this lesson

You will learn:

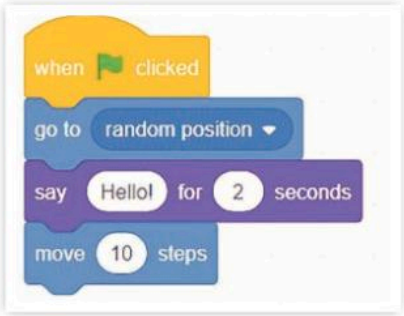
- how to make changes to a program
- how to find and fix errors.

Commands to match actions

A student wanted to make a program. They decided the sprite would:

- jump to a random place
- say "Hello!"
- move around the screen forever.

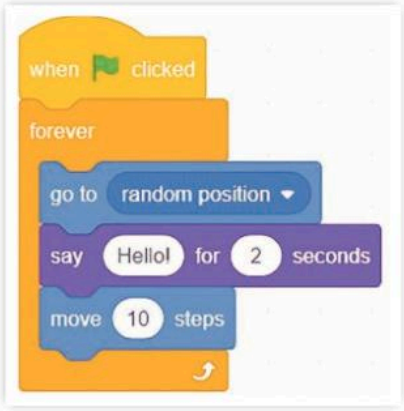
The student found blocks to match the actions. Here is the program made of those blocks.



The student ran the program. The program did not work the way they wanted. The sprite moved just once, then stopped.

Using a loop

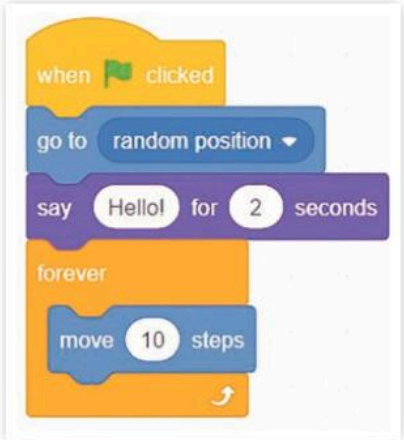
The student decided to use a 'forever loop'. They put all the command blocks inside the loop. Here is the program they made.



The student ran the program. The program did not work the way they wanted. Too many blocks are inside the loop. The sprite keeps jumping to a new place. The sprite keeps saying "Hello!"

Inside and outside the loop

Some commands should go before the loop. Those are the commands that must happen only once.



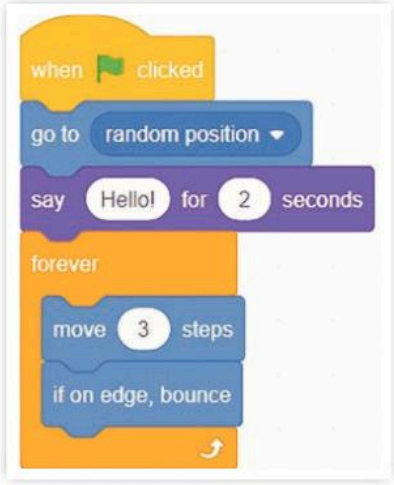
Some commands should be put inside the loop. Those are the commands that will be repeated.

Look at the final picture on the previous page. You will see the program with all the commands in the right place.

Add some bounce

The sprite moves until it is almost off the screen and then stops. You must add a block that tells the sprite to 'bounce' if it hits the side.

The finished program is shown here.





Activity

Using the ideas on this page make a program that makes a butterfly move about the screen.



Extra challenge

Look in the program you made to find the block that says 'move 10 steps'.

Change the number. Run the program and see what is different.

Explore what happens when you use different numbers.



Explore more

Make a new program. Use a sprite that looks like a ball. Make it bounce around the screen.

Add different types of ball. Set them to different speeds.

